S-JVF02 SMT stick vibration feeder — Factory Inspection Checklist

JUKI RS-1 • English

DRAFT – contains suggested or TBD items, not for formal release

Shenzhen Southern Machinery Sales and Service Co., Ltd.
smthelp.com • info@smthelp.com • WhatsApp +86 136 0256 2576
SM-FQC-S-JVF02 • Rev. A • template 1.0 • smt-brand-kit

Model	S-JVF02	**Machine / config**	JUKI RS-1
Serial No.		**Customer**	
Order / PI No.		**Destination / voltage**	
Inspector		**Date**	

No.	Grade	Check item	Acceptance criterion	Nominal / LSL ~ USL	Method / tool	Sampling	Source	Result	Measured	Conclusion / remark	Inspector	Date
A Appearance & Structure												
A01	MI	Surface finish	No scratches, dents, deformation or rust; finish even, no peeling; scratches on non-visible faces ⩽ 5 mm and ⩽ 2	— ~ 5 mm ★	Visual, lighting ⩾ 500 lx, distance 30 - 50 cm	100%	General G-APP-01	☐ Pass ☐ Fail ☐ NA				
A02	MI	Overall dimensions	365 × 66 × 125 mm, tolerance ± 1 mm	365×66×125 / -1 ~ +1 mm ★	Steel rule / caliper, measure L × W × H	2 units per lot	Product page § 03 spec table 'Overall size'	☐ Pass ☐ Fail ☐ NA				
A03	MI	Weight	6.5 kg, tolerance ± 0.2 kg	6.5 / 6.3 ~ 6.7 kg ★	Electronic scale, without tubes	2 units per lot	Product page § 03 spec table 'Weight'	☐ Pass ☐ Fail ☐ NA				
A04	MA	Nameplate & marking	Model S-JVF02, serial no., 24 V DC / 15 W legible and fixed; Southern Machinery marking only, no other maker's marks	—	Visual; wipe 10 times with lint-free cloth, no fading	100%	Product page § 03 spec table; smt-brand-kit brand rules	☐ Pass ☐ Fail ☐ NA				
A05	MA	Structural completeness	Main body, top plate, front cover, guide rail, support frame, handlebar and control unit present, matching the 14-item list in Manual Ch. 9	—	Check item by item against parts S-JVF02-052801 ~ 052814	100%	Manual § 9 p.12 parts list	☐ Pass ☐ Fail ☐ NA				
A06	MI	Cleanliness	Lanes, pick-up area and interior free of metal chips, oil and foreign matter	—	Visual + lint-free cloth wipe	100%	Manual § 7 p.10 maintenance	☐ Pass ☐ Fail ☐ NA				
B Mechanical Assembly & Fastening												
B01	MA	Screw fastening	No missing or stripped screws; all tightened with torque-seal mark; M3 torque 0.5 - 0.7 N・m	0.5 ~ 0.7 N・m ★	Torque screwdriver, sample ⩾ 5 screws; visual check of marks	⩾5 screws per unit	General G-MEC-01	☐ Pass ☐ Fail ☐ NA				
B02	MA	Handlebar action	Handlebar raises and lowers smoothly, no binding; returns and locks when released	—	Operate by hand 10 times	100%	Manual § 4.1 p.7–8	☐ Pass ☐ Fail ☐ NA				
B03	MA	Locating pin	Locating pin (S-JVF02-052807) not loose or bent; enters the feeder bank alignment hole smoothly	—	Hand check + installation test	100%	Manual § 4.1 p.7; § 9 p.12 #12	☐ Pass ☐ Fail ☐ NA				
B04	MA	Connector	Connector (S-JVF02-052814) pins not bent, recessed or oxidised; firmly mounted	—	Visual with magnifier	100%	Manual § 4.1 p.8; § 9 p.12 #7	☐ Pass ☐ Fail ☐ NA				

No.	Grade	Check item	Acceptance criterion	Nominal / LSL ~ USL	Method / tool	Sampling	Source	Result	Measured	Conclusion / remark	Inspector	Date
B05	MA	Install / remove	Install in JUKI RS-1 feeder bank (or equivalent test bank) per Manual 4.1 / 4.2: smooth insertion, no perceptible play once locked (⩽ 0.2 mm); smooth removal	— ~ 0.2 mm ★	Install / remove 5 times; dial gauge or hand check	100%	Manual § 4.1–4.2 p.7–8	☐ Pass ☐ Fail ☐ NA				
B06	MI	Spring	Spring (S-JVF02-052806) not deformed, rebounds normally	—	Visual + hand press	100%	Manual § 9 p.12 #4	☐ Pass ☐ Fail ☐ NA				
B07	MI	Slot interchangeability	Installs and locks in 2 different feeder bank slots	—	Installation test	2 units per lot	General G-MEC-02	☐ Pass ☐ Fail ☐ NA				
C Feeder Feeding & Vibration / Belt Mechanism												
C01	MA	Lanes & guide rail	4 independent lanes present; guide rail (S-JVF02-052810) parallel, no burrs or deformation; 66 mm tube fits each lane smoothly	4 / 66 lanes / mm	Visual; trial-fit a standard 66 mm tube in each lane	100%	Product page § 03 'Input channels / Stick width'; Manual § 9 #13	☐ Pass ☐ Fail ☐ NA				
C02	MA	Three-point tube fixing	Locating ring (rear), leaf spring (middle) and locating piece (front) hold the tube securely; leaf spring presses down on the tube	—	Push the fitted tube by hand: no movement	100%	Manual § 5.1 p.9	☐ Pass ☐ Fail ☐ NA				
C03	MA	Vibrator	Vibrator (S-JVF02-052809) firmly mounted; even vibration when powered, no abnormal or knocking noise	—	Power on, listen and feel	100%	Manual § 9 p.12 #11	☐ Pass ☐ Fail ☐ NA				
C04	MA	Power knob	Turns smoothly through full travel; vibration strength adjusts continuously from min to max	—	Turn from min to max, observe amplitude	100%	Manual § 5.2 p.9	☐ Pass ☐ Fail ☐ NA				
C05	MA	Frequency knob	Turns smoothly through full travel; vibration frequency adjusts continuously, a smooth-flow point can be found	—	Turn from min to max, observe component flow	100%	Manual § 5.2 p.9	☐ Pass ☐ Fail ☐ NA				
C06	MA	Feeding test (small parts)	3 × 3 mm parts: after tuning, parts move smoothly and continuously to the pick-up position, no jam, jump or flip; ⩾ 100 pcs per lane	100 ~ — pcs/lane ★	Load a tube in each lane, count and observe	100%	Product page § 03 'Component size range'; qty G-RUN-01	☐ Pass ☐ Fail ☐ NA				
C07	MA	Feeding test (large parts)	30 × 10 mm parts: as above, ⩾ 50 pcs per lane	50 ~ — pcs/lane ★	Load a tube in each lane, count and observe	100%	Product page § 03 'Component size range'; qty G-RUN-01	☐ Pass ☐ Fail ☐ NA				
C08	MA	Pick-up position	Tube top face cut at the nozzle pick-up area; parts stop at a consistent height and attitude	— ~ 0.2 mm ★	Visual / feeler gauge, height difference ⩽ 0.2 mm ★	100%	Manual § 5.1 p.9; § 8 p.11	☐ Pass ☐ Fail ☐ NA				
C09	MA	Lane independence	Each of the 4 lanes feeds independently; emptying or changing one lane does not disturb the others	—	Remove tubes one by one, observe other lanes	100%	Product page § 03 'Input channels'	☐ Pass ☐ Fail ☐ NA				
C10	MA	Continuous run	All 4 lanes loaded, continuous run ⩾ 2 h: no jams, no vibration stop; housing temperature rise ⩽ 20 K	⩾ 2 h; ΔT ⩽ 20 K ★	Timer; IR thermometer on housing near vibrator	100%	General G-RUN-02	☐ Pass ☐ Fail ☐ NA				

No.	Grade	Check item	Acceptance criterion	Nominal / LSL ~ USL	Method / tool	Sampling	Source	Result	Measured	Conclusion / remark	Inspector	Date
C11	MI	Running noise	At normal setting, measured 1 m from the feeder: ⩽ 70 dB(A)	— ~ 70 dB(A) ★	Sound level meter (A-weighted), background ⩾ 10 dB below reading	1 unit per lot	General G-RUN-03	☐ Pass ☐ Fail ☐ NA				
C12	MA	Pick test on machine (if available)	On JUKI RS-1, ⩾ 500 picks, pick success ⩾ 99.5 %	99.5 ~ — % ★	Read machine pick / rejection log	First unit per lot (if RS-1 available)	General G-RUN-04	☐ Pass ☐ Fail ☐ NA				
C13	MI	Belt mechanism	S-JVF02 is vibration feeding with no belt mechanism; record NA	—	Check construction	—	Manual § 2.1 p.4 (vibration type)	☐ Pass ☐ Fail ☐ NA				
D Sensors & Photoelectric Detection												
D01	MA	Pilot lamp	Lamp on when the feeder is switched on and locked; off when switched off or not locked	—	Operate switch / handlebar with power on	100%	Manual § 2.2 p.5; § 8 p.11	☐ Pass ☐ Fail ☐ NA				
D02	MA	Power switch	Switch on/off positive; vibration stops immediately when off	—	Operate 10 times	100%	Manual § 1 p.3; § 2.2 p.5	☐ Pass ☐ Fail ☐ NA				
D03	MA	Feeder recognition / communication	After installation, RS-1 recognises the feeder with no feeder or connection alarm	—	Install on machine, check machine screen	100%	Product page § 03 'Interface'	☐ Pass ☐ Fail ☐ NA				
D04	MA	Photoelectric / part-presence sensor	**TBD: Not listed in Manual; if fitted on this batch, signal and indication correct when blocked / part present; NA if not fitted**	**TBD**	Block with shutter / part	100%	Not in sources	☐ Pass ☐ Fail ☐ NA				
D05	MI	Internal wiring	Neatly routed and tied, not pinched or damaged, clear of vibration parts	—	Open front cover, visual	100%	General G-ELE-01	☐ Pass ☐ Fail ☐ NA				
E Electrical & Safety (Grounding, Insulation, E-stop)												
E01	CR	Supply voltage	24 V DC at connector, allowed 21.6 - 26.4 V (± 10 %)	24 / 21.6 ~ 26.4 V DC ★	Multimeter at connector power pins	100%	Product page § 03 'Voltage'; tolerance G-ELE-02	☐ Pass ☐ Fail ☐ NA				
E02	MA	Current / power	Rated 15 W; at 24 V and max vibration setting current ⩽ 0.65 A (≈ 15.6 W)	15 W / — ~ 0.65 A ★	DC clamp meter / bench supply reading	100%	Product page § 03 'Power'; current derived from 15 W/24 V	☐ Pass ☐ Fail ☐ NA				
E03	CR	Power-on check	Polarity matches connector pin-out; no smoke, smell or abnormal heat at power-on	—	Power on and observe 5 min	100%	General G-ELE-03	☐ Pass ☐ Fail ☐ NA				
E04	CR	Earth continuity	Resistance between metal body and connector ground pin ⩽ 0.1 Ω	— ~ 0.1 Ω ★	Low-resistance / micro-ohm meter	100%	General G-ELE-04	☐ Pass ☐ Fail ☐ NA				
E05	CR	Insulation resistance	Power pins (shorted) to metal housing ⩾ 1 MΩ at 250 V DC (ref. IEC 60204-1, 24 V circuit)	1 ~ — MΩ @250 V DC ★	Insulation tester, control unit electronics isolated	100%	General G-ELE-05 (ref. IEC 60204-1)	☐ Pass ☐ Fail ☐ NA				
E06	CR	E-stop interaction	Feeder has no own E-stop; with RS-1 E-stop pressed, machine stops and feeder vibration stops or is safe	—	Press machine E-stop on test	First unit per lot	General G-SAF-01	☐ Pass ☐ Fail ☐ NA				

No.	Grade	Check item	Acceptance criterion	Nominal / LSL ~ USL	Method / tool	Sampling	Source	Result	Measured	Conclusion / remark	Inspector	Date
E07	CR	Switch-off before handling	Switch off before removing feeder or changing tube (Manual safety section); no vibration when off	—	Verify per Manual	100%	Manual § 1 p.3; § 4.2 p.8	☐ Pass ☐ Fail ☐ NA				
E08	MA	Safety label	**TBD: "Keep hands clear of pick-up area" or equivalent label on the body or near the lanes**	**TBD**	Visual	100%	Manual § 1 p.3 (safety notes)	☐ Pass ☐ Fail ☐ NA				
E09	MA	Sharp edges & pinch points	No sharp edges or burrs; no pinch risk at handlebar or pick-up area	—	Glove touch check	100%	General G-SAF-02	☐ Pass ☐ Fail ☐ NA				
F Specification Verification												
F01	MA	Model	S-JVF02 SMT stick (tube) vibration feeder	S-JVF02	Check nameplate vs order	100%	Manual cover p.1; § 3 p.6	☐ Pass ☐ Fail ☐ NA				
F02	CR	Compatible machine	JUKI RS-1 feeder bank	JUKI RS-1	Check order / install test	100%	Product page § 03 'Model compatibility'	☐ Pass ☐ Fail ☐ NA				
F03	CR	Voltage	24 V DC	24 V DC	Nameplate + E01 reading	100%	Product page § 03 'Voltage'	☐ Pass ☐ Fail ☐ NA				
F04	MA	Power	15 W	15 W	Nameplate + E02 reading	100%	Product page § 03 'Power'	☐ Pass ☐ Fail ☐ NA				
F05	MA	Frequency	**TBD: DC supply, no mains frequency requirement; vibration frequency adjustable by knob (range TBD)**	**TBD**	Check + C05 result	100%	Manual § 3 p.6 (no range given)	☐ Pass ☐ Fail ☐ NA				
F06	MA	Component size range	3 × 3 mm - 30 × 10 mm	3×3 – 30×10 mm	Check C06 / C07 results	100%	Product page § 03 'Component size range'	☐ Pass ☐ Fail ☐ NA				
F07	MA	Tube width / lanes	66 mm tube; 4 independent lanes	66 / 4 mm / lanes	Check C01 result	100%	Product page § 03 'Stick width / Input channels'	☐ Pass ☐ Fail ☐ NA				
F08	MA	Documents & packing	English Operation Manual (Rev. 2025), conformity certificate, this checklist and packing list included; anti-static bag + foam, handlebar and lanes secured	—	Check against packing list	100%	Manual cover (Rev. 2025); packing G-PAK-01	☐ Pass ☐ Fail ☐ NA				

Items		**Pass**		**Fail**		**NA**		**Overall ☐ Release ☐ Concession ☐ Hold / rework**

Inspector signature / date	**QC manager signature / date**	**Customer witness signature / date**

Sampling & judgement rules: Machines / single units: every item, 100 % per unit. Lots of identical small units (feeders, parts): items marked 'per lot' sampled to ISO 2859-1 / GB/T 2828.1 general level II; CR items 100 %; MA AQL 0.65; MI AQL 1.5. 1) Any CR (critical) item fails: unit / lot not released; fix and re-inspect all items. 2) MA (major) item fails: rework, re-inspect that item and related items; release only when all pass. 3) Only MI (minor) items fail and ⩽ 2: QC manager may sign a concession release with the disposition in the remark. 4) NA needs a reason in the remark; a sheet with ★ suggested values or TBD items is a draft and cannot be used for formal release.

Grade: CR critical / MA major / MI minor. ★ = suggested value (not in the sources, proposed from general practice, to be confirmed); 'TBD' = data missing, no value entered. Source: manual section and page / product page section / video time / general-practice code. Terms per glossary.